Rajagopal Somaskandan

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Professional Summary

Entry-level Machine Learning Engineer with hands-on experience building and deploying end-to-end AI systems using FastAPI and modern ML frameworks. Developed production-oriented projects including fraud detection (94% recall, ROC-AUC 0.995) and real-time ML pipelines. Strong focus on backend systems, model explainability, and scalable AI deployment.

Technical Skills

Languages: Python, Java, SQL
Machine Learning: Scikit-learn, XGBoost, LightGBM, PyTorch, SHAP
AI / NLP Systems: BERT, Sentence-Transformers, LangChain, ChromaDB, FAISS, RAG
Backend & APIs: FastAPI, Flask, REST APIs, Model Deployment
Computer Vision & 3D: OpenCV, Optical Flow, COLMAP, 3D Gaussian Splatting
Databases: PostgreSQL, MongoDB, Redis
Tools & Platforms: Docker, ngrok, Git, GitHub, FFmpeg, Streamlit, Render, Netlify
Data Analysis: Pandas, NumPy, Matplotlib, Power BI

Education

Sathyabama Institute of Science and Technology — **B.Sc. Computer Science (AI Specialization)** | First Class — *2025*
Sathyabama Institute of Science and Technology — **M.Sc. Computer Science** | — *Expected 2027*

Experience

Production AI Developer (Intern) | *InnoXR Labs Pvt. Ltd.* — *Mar 2026 – Jun 2026*
- Built and deployed ML-powered FastAPI microservices for real-time inference in AR/immersive applications.
- Designed API-first ML systems integrating model inference with scalable backend endpoints.
- Delivered end-to-end features from model development to deployment in a fast-paced startup environment.

Data Analyst Intern | *India Advocacy* — *Jul 2024 – Oct 2024*
- Built Java + Selenium pipelines to scrape and process **1M+ records**, transforming raw data into **15–20k structured datasets** through multi-stage validation.

Security Trooper (National Service) | *Singapore Armed Forces* — *Jan 2020 – Jan 2022*
- Executed security operations in regulated environments, ensuring strict protocol compliance and operational readiness.
- Supported COVID-19 response initiatives through structured data collection and reporting under time-sensitive conditions.

Sales Associate (Part-Time) | *Bata* — *Feb 2022 – Aug 2022*
- Handled high-volume customer interactions in a fast-paced retail environment, ensuring efficient service delivery.
- Managed billing, inventory coordination, and issue resolution to support smooth store operations.

Projects

FinGuard Pro: Real-Time Financial Fraud Detection & Compliance System — *IEEE Research Project (ICAAIC 2025)* — *2025*
XGBoost, SHAP, FastAPI, Streamlit, SQL
- Built a real-time fraud detection platform with REST APIs and role-based dashboards for transaction monitoring.
- Achieved **94% recall** and **ROC-AUC 0.995** on imbalanced financial data; integrated SHAP for explainable, audit-ready decisions.
- Developed automated reporting pipelines (PDF/ZIP) to support scalable fraud investigation workflows.

CareerGenie: AI Career Intelligence Platform (RAG + LTR + Multi-Agent) — *Hackathon Winner (Origin 2026)* — *2026*
FastAPI, React, Ollama (LLaMA 3.2), ChromaDB, Sentence-Transformers
- Built an AI career platform integrating resume parsing, ATS scoring, semantic job matching, and mock interview generation.
- Implemented RAG-based retrieval using MiniLM embeddings + ChromaDB, enabling efficient semantic job matching.
- Developed adaptive ranking using **BPR learning-to-rank** with feedback simulation for personalised job recommendations.
- Designed a multi-agent validation system (propose–critique–synthesise) to improve output reliability and reduce hallucinations.

MonoSplat: Single-Video 3D Reconstruction Pipeline — *Internship Project* — *2026*
PyTorch, FastAPI, COLMAP, FFmpeg
- Built a single-video 3D reconstruction pipeline converting RGB video into interactive Gaussian Splat scenes viewable in browser.
- Implemented SfM-based camera pose estimation (COLMAP) and GPU-based Gaussian Splat training pipeline.
- Designed hybrid CPU–GPU architecture with local preprocessing (FFmpeg, SfM) and cloud GPU training (Colab).
- Developed FastAPI backend with async job pipeline and Three.js viewer achieving **30+ FPS** real-time rendering.

S.A.F.E: Real-Time Crowd Anomaly & Risk Detection — *Design Thinking Project* *2026*

OpenCV, Farneback Optical Flow, MAD, FastAPI, Streamlit

- Developed a privacy-preserving anomaly detection system using optical flow aggregation for real-time CCTV monitoring.
- Reduced false positives by **40–60%** via global motion compensation.
- Achieved **ROC-AUC 0.8391** on cross-domain dataset without retraining, improving generalization.

Redline: Multilingual Preventive Healthcare Chatbot — *Smart India Hackathon (SIH 2025)* *2026*

IndicBERT, Groq LLaMA 3.3, FastAPI, Telegram API

- Built a multilingual healthcare chatbot combining IndicBERT with LLM fallback for intent classification.
- Achieved **82.4% intent accuracy** and **95.6% safety filtering accuracy** using rule-based validation.
- Deployed real-time alert system with region-specific health data via Telegram API.

SourceUp: Explainable Supplier Recommendation System — *Final Year Capstone* *2026*

Sentence-BERT, FAISS, FastAPI, Scikit-learn, Selenium, Java

- Built a supplier intelligence pipeline using Java + Selenium for large-scale web data collection (**1M+ records**).
- Implemented semantic search using Sentence-BERT + FAISS for natural language supplier queries.
- Designed constraint-based scoring model incorporating price, delivery time, and reliability.
- Added explainability layer providing feature-level insights and risk classification for decision support.

Publications & Conferences

FinGuard Pro: Explainable AI for Financial Fraud Detection | *ICAAIC 2025* *ISBN: 979-8-3315-6587-9*

Real-Time Explainable Fake News Detection using BERT-LSTM | *ICASET 2025*

Achievements & Hackathons

Industry Innovation Hackathon | *Sathyabama Institute of Science and Technology — National Level* *Mar 2026*

- Won ***Best Industry Innovation Award*** at Industry Innovation ’26 (organized with ATRIBS and Bluewind Innovations) — defeating 63+ teams from institutions across India with the CareerGenie AI platform.

Origin 24-Hour Hackathon | *SIMATS Engineering, Chennai* *Apr 2026*

- Competed in a 24-hour sprint organized by the SIMATS Hackathon Club, Saveetha Institute of Medical and Technical Sciences.